

GEOS 451 – Data Capture 2

Final Project

*Mapping Wildfire Burn Severity for the Blackrock Mountain (Devil's Head) Fire Using
dNBR Analysis of Sentinel-2 Imagery*

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1. Introduction

In September 2020, a wildfire ignited near the base of Blackrock Mountain in the North Ghost River watershed of the Bighorn Backcountry, Alberta, reportedly caused by an abandoned campfire (The Globe and Mail, 2020). Officially named the Devil's Head wildfire, the fire grew rapidly under dry, high-wind conditions before being classified as held in mid-October 2020, ultimately burning approximately 2,420 hectares (CBC News, 2021).

The purpose of this project is to (1) accurately delineate the boundary of the Devil's Head wildfire within the defined study area, (2) determine the total area burned, and (3) classify the burned area into low, medium, and high severity classes. The method selected for this analysis is the differenced Normalized Burn Ratio (dNBR), a widely used remote sensing technique for quantifying wildfire severity from multispectral satellite imagery (Wassner & Cattau, 2020).

2. Imagery Data

Sentinel-2 Level-2A (surface reflectance) imagery was selected for this analysis over Landsat 8 due to its finer spatial resolution (10 m for the near-infrared band and 20 m for the shortwave-infrared band, versus 30 m for Landsat 8) and shorter revisit interval (5 days versus 16 days), both of which improve the precision of the fire boundary and severity classification for a burn extent of this size. Imagery was acquired free of charge through the Copernicus Data Space Ecosystem browser.

Higher-resolution commercial imagery from Planet (PlanetScope, 3 m resolution) was also considered for this analysis. It was not selected for two reasons. First, PlanetScope's standard 4-band product does not include a shortwave-infrared (SWIR) band, which is required to calculate NBR; using it would have required sourcing a different Planet product or supplementary SWIR data. Second, and more importantly, finer spatial resolution increases sensitivity to fine-scale spectral noise, such as individual tree shadows, small rock outcrops, and pixel-level reflectance variation, that can be misclassified as burned when it is not, producing more false-positive speckling in the severity classification than a coarser sensor would. This project already required a manual noise-removal step (Section 3.5) to clean scattered classification artifacts from 10-20 m Sentinel-2 data; a 3 m product would likely have introduced substantially more speckle, requiring a larger and more aggressive cleanup step that risks removing genuinely small burned patches along with noise. Sentinel-2 was judged to offer resolution fine enough to resolve the fire's boundary and severity pattern at the scale of this project (2,478 ha), while keeping this noise-versus-detail trade-off manageable.

2.1 Pre-Fire Image

- Acquisition date: August 24, 2020
- Platform / sensor: Sentinel-2 (MSI – MultiSpectral Instrument), Level-2A
- Bands used: B08 (NIR, 10 m native resolution), B12 (SWIR, 20 m native resolution)
- Format: TIFF, 16-bit, UTM Zone 11N (EPSG:32611)
- Cloud cover: 0% over the study area (tile-reported cloud cover 30%, but the study area itself was visually confirmed clear)

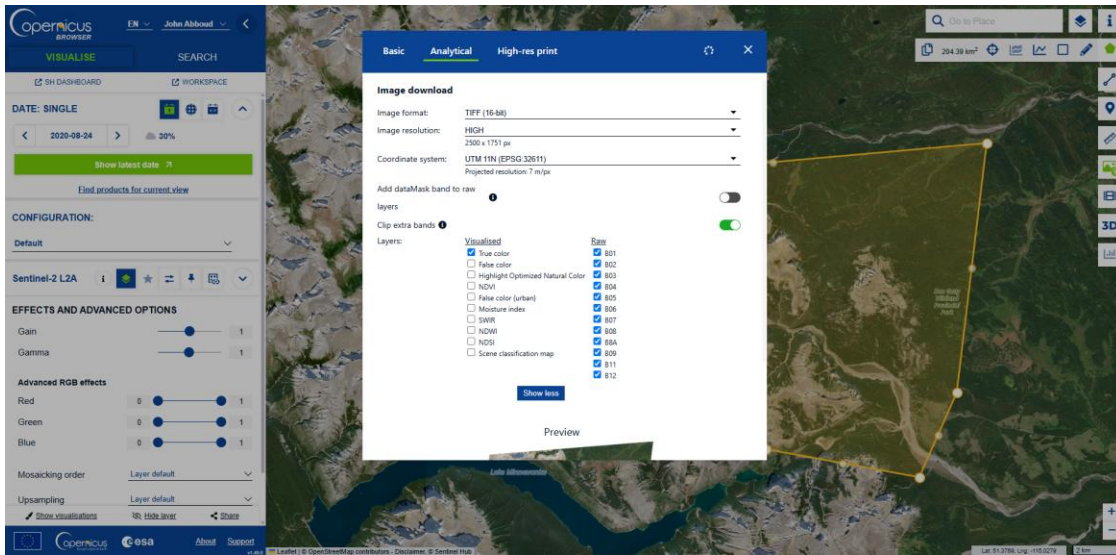


Figure 1. Copernicus Data Space Ecosystem download configuration for the August 24, 2020 pre-fire Sentinel-2 scene.



Figure 2. Study area boundary overlaid on the August 24, 2020 pre-fire true-colour image, confirming the absence of cloud cover.

2.2 Post-Fire Image

Selecting a suitable post-fire image required more consideration than the pre-fire image. The Devil's Head fire remained active and out of control until it was classified as held in mid-October 2020 (CBC News, 2021). A candidate Sentinel-2 scene from October 3, 2020 was rejected after closer inspection revealed active smoke over the study area (Figure 3), confirming the fire had not yet been contained. A personal field observation further corroborated this: the author rode the nearby Lake Minnewanka trails on October 11, 2020 (Strava activity titled "Under Snow Flurries") and photographed early season snowfall at similar elevations within the surrounding area. No further cloud-free Sentinel-2 scenes were

available in November 2020, and December 2020 imagery was fully snow-covered, ruling out a same-season, near-term post-fire image.

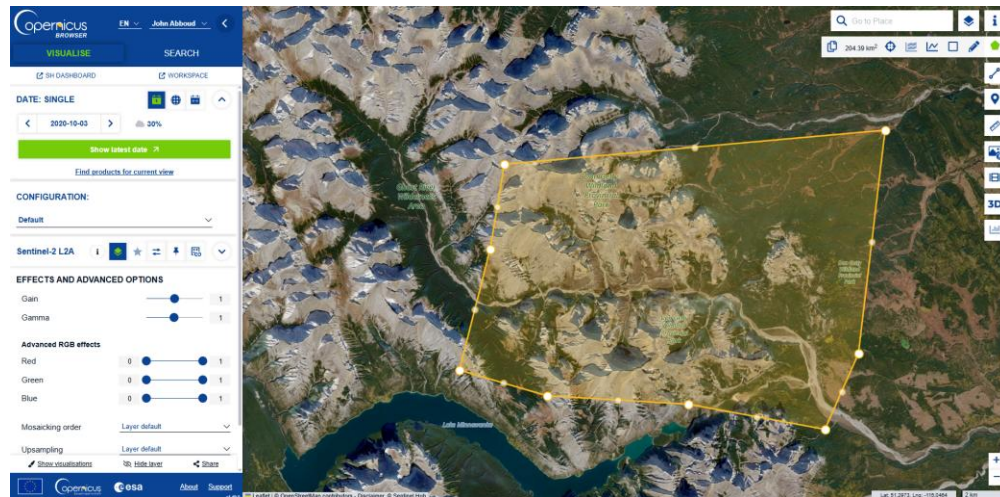


Figure 3. October 3, 2020 Sentinel-2 scene showing active smoke over the study area, indicating the fire was not yet contained.

A May 2021 scene was initially considered, but comparison against the August 2020 pre-fire image revealed a phenological mismatch: large portions of the study area that were not burned appeared dramatically different between the two dates simply because spring imagery captures dormant, post-snowmelt grasses and shrubs rather than full summer vegetation. Using a phenologically mismatched image pair risks introducing false positives into the dNBR result, since the seasonal vegetation change is confounded with true fire-related change across areas that never burned.

To avoid this, a same-season, one-year-anniversary image was instead selected for August 29, 2021. This approach, comparing imagery from equivalent points in the growing season one year apart, is a recognized alternative when weather, smoke, or snow prevent a near-term post-fire capture, and it preserves phenological comparability between the two dates at the cost of allowing roughly one year of vegetation recovery within the burn scar.

- Acquisition date: August 29, 2021
- Platform / sensor: Sentinel-2 (MSI), Level-2A
- Bands used: B08 (NIR, 10 m native resolution), B12 (SWIR, 20 m native resolution)
- Format: TIFF, 16-bit, UTM Zone 11N (EPSG:32611)
- Cloud cover: 0% over the study area



Figure 4. August 29, 2021 Sentinel-2 scene selected as the post-fire image, showing the burn scar as a distinct darker patch within the study area, consistent in location and shape with the scar visible in intermediate spring 2021 scenes.

3. Methodology

All geoprocessing was performed in ArcGIS Pro using the Spatial Analyst extension. The overall workflow followed the standard dNBR method: (1) calculate the Normalized Burn Ratio (NBR) for the pre-fire and post-fire images, (2) difference the two NBR rasters to produce dNBR, (3) reclassify dNBR into severity classes, (4) convert the classified raster to polygons, (5) clip to the study area and remove classification noise, and (6) dissolve the burned classes into a single wildfire boundary feature.

3.1 Normalized Burn Ratio (NBR) Calculation

NBR was calculated separately for each date using the standard formula:

$$NBR = (NIR - SWIR) / (NIR + SWIR)$$

For Sentinel-2, NIR corresponds to Band 8 and SWIR corresponds to Band 12. The calculation was performed in Raster Calculator, wrapping each band in the Float() function to force floating-point division (the raw bands are stored as integers, which would otherwise truncate the result):

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(Float(Raster("B08")) - Float(Raster("B12"))) / (Float(Raster("B08")) + Float(Raster("B12")))
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This produced two rasters, NBR_pre (from the August 24, 2020 image) and NBR_post (from the August 29, 2021 image), each with a valid value range of approximately -0.93 to 0.82.

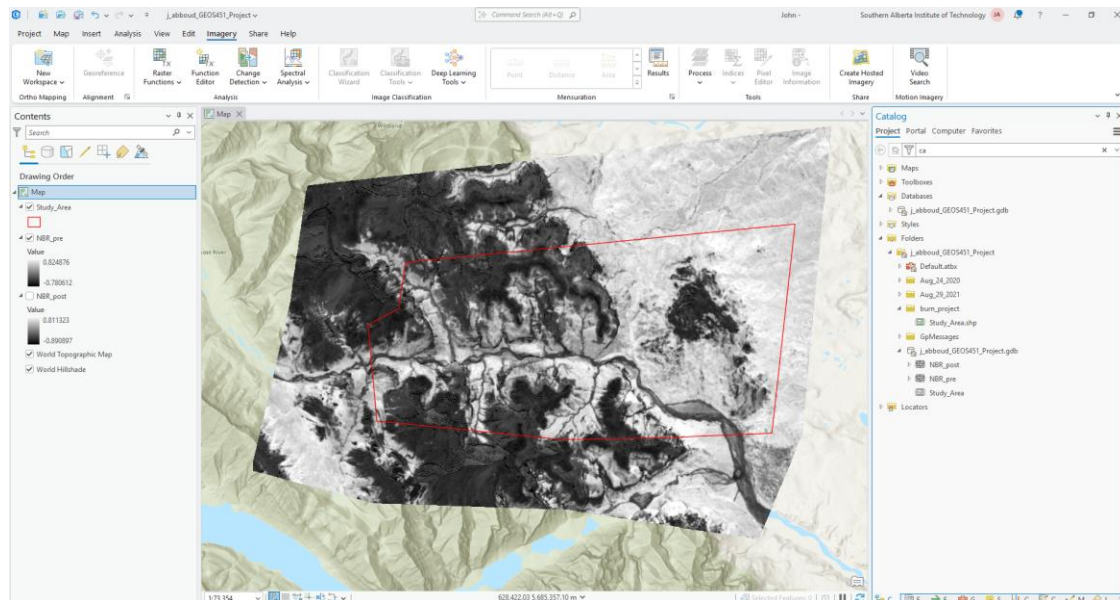


Figure 5. NBR_pre raster displayed in ArcGIS Pro, clipped to the general image extent, with the study area boundary shown in red.

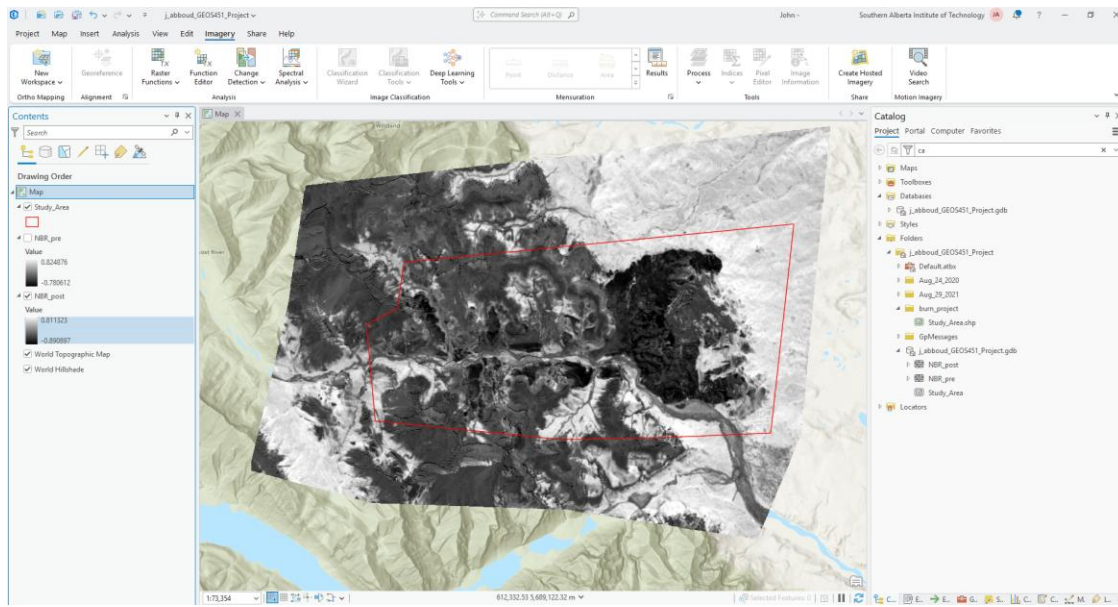


Figure 6. *NBR_post* raster displayed in ArcGIS Pro.

3.2 Differenced NBR (dNBR)

The dNBR raster was calculated by subtracting the post-fire NBR from the pre-fire NBR:

$$dNBR = NBR_{pre} - NBR_{post}$$

This ordering (pre minus post) follows the standard convention so that burned areas, which lose reflectance in the near-infrared and gain reflectance in the shortwave-infrared, produce positive dNBR values, consistent with the USGS/Key and Benson severity thresholds applied in Section 3.3. The resulting dNBR raster had a value range of approximately -0.93 to 1.30 and clearly highlighted the burn scar as a distinct area of high positive values against a background of near-zero values elsewhere in the study area (Figure 7).

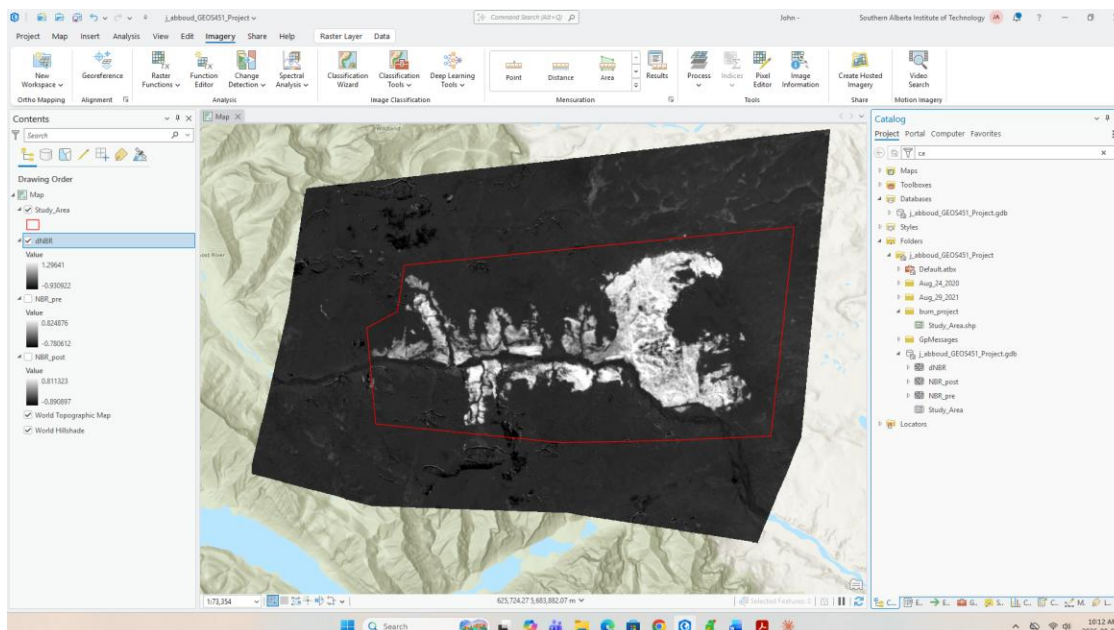


Figure 7. dNBR raster (NBR_{pre} minus NBR_{post}), with the burn scar visible as a bright patch within the study area boundary.

3.3 Severity Classification

The dNBR raster was reclassified into four categories using the Reclassify tool. The standard six-class USGS/Key and Benson (2006) dNBR severity scheme was collapsed into four classes (unburned, low, medium, high) to match the project's required deliverable of low/medium/high severity plus an unburned background class. The 'enhanced regrowth' class from the standard scheme was merged into unburned, and the standard 'moderate-low' and 'moderate-high' classes were merged into a single medium severity class.

dNBR range	Reclassified value	Severity class
-0.93 to 0.10	1	Unburned
0.10 to 0.27	2	Low severity
0.27 to 0.66	3	Medium severity
0.66 to 1.30	4	High severity

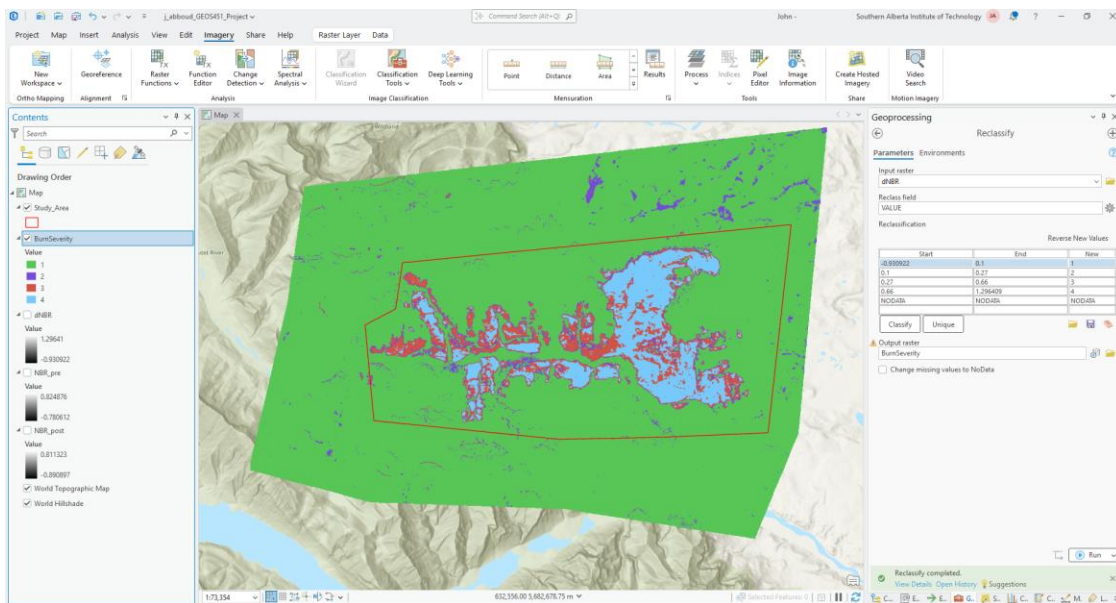


Figure 8. Reclassify tool output (BurnSeverity raster), showing the four-class severity scheme applied to the study area.

3.4 Conversion to Feature Classes

The classified BurnSeverity raster was converted to polygons using the Raster to Polygon tool (Figure 9), producing BurnSeverity_poly. This layer was then clipped to the Study_Area boundary using the Clip tool, producing BurnSeverity_Clip (Figure 10).

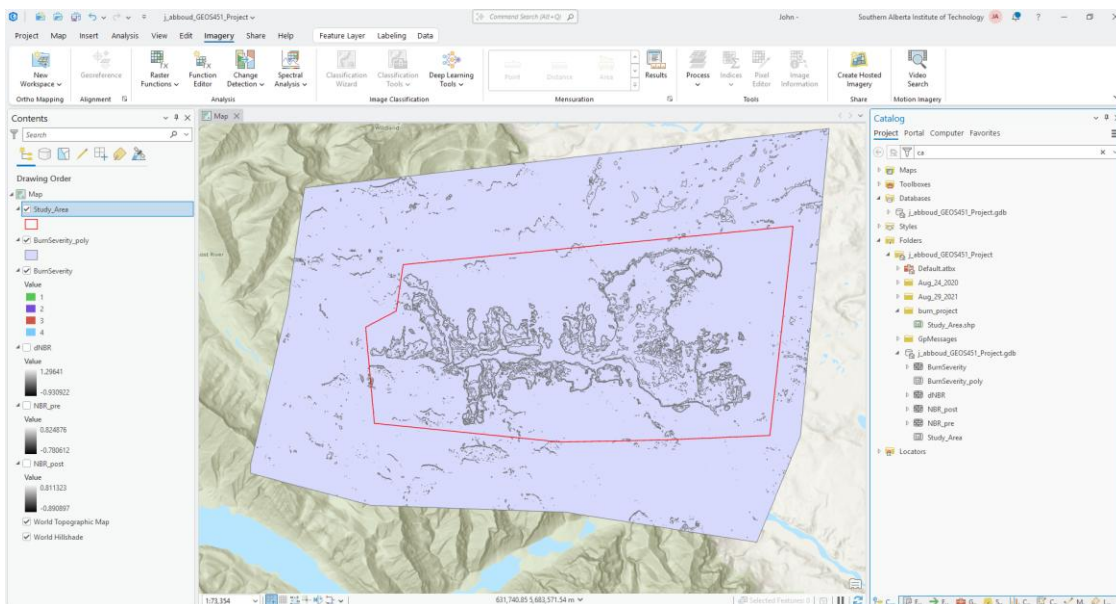


Figure 9. Raw output of the Raster to Polygon conversion prior to clipping.

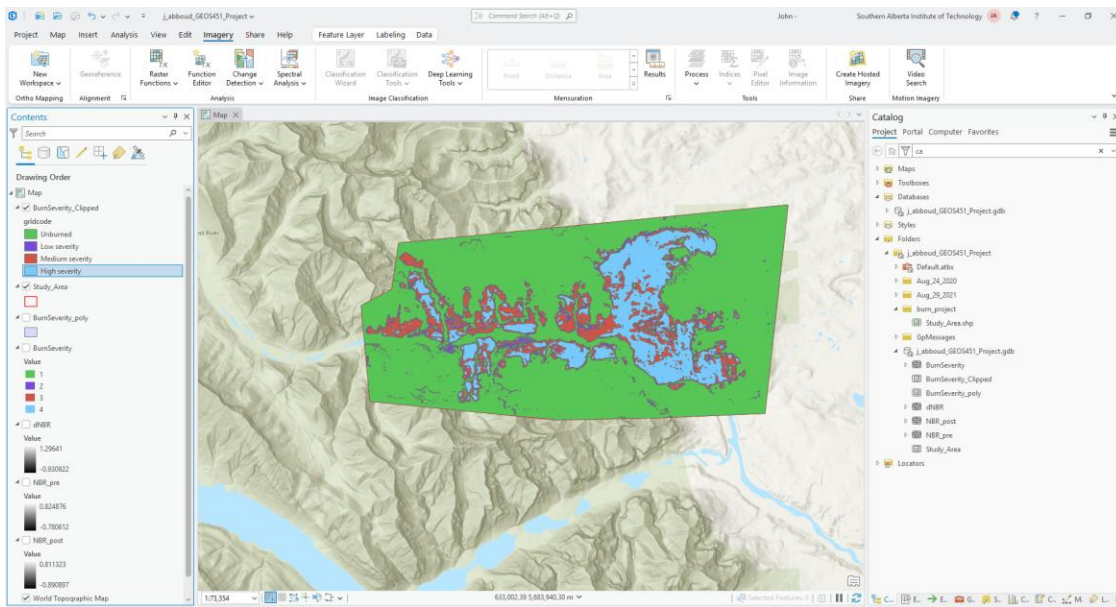


Figure 10. BurnSeverity_Clippped layer, clipped to the Study_Area boundary and symbolized by severity class.

3.5 Noise Removal

Per-pixel raster classification typically introduces small, scattered polygons ("salt-and-pepper" noise) caused by minor spectral anomalies such as water edges, shadow, or isolated noisy pixels crossing a class threshold. To remove this noise without leaving gaps in the polygon layer, features smaller than 400 m² (approximately four Sentinel-2 pixels) were selected using Select By Attributes (Shape_Area < 400) and merged into their largest bordering neighbour using the Eliminate tool. This removed 2,660 small polygons, producing the cleaned layer BurnSeverity_Clippped_Cleaned, which serves as the project's final burn severity feature class.

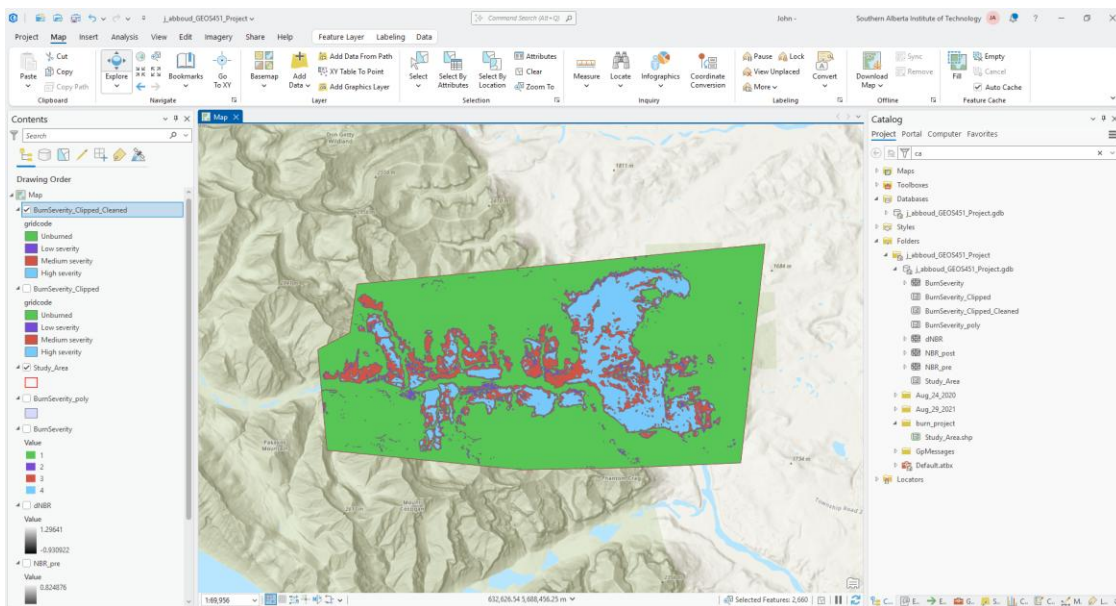


Figure 11. BurnSeverity_Clippped_Cleaned after removing 2,660 small noise polygons with the Eliminate tool. Compare against Figure 10 to note the reduction in scattered speckling within the unburned background.

3.6 Wildfire Boundary Delineation

The overall wildfire boundary was derived by selecting all polygons in BurnSeverity_Clippped_Cleaned with a gridcode not equal to 1 (i.e., all low, medium, and high severity polygons, excluding unburned), then dissolving the selection into a single multipart feature using the Dissolve tool (Figure 12), with "Create multipart features" enabled so that any disconnected burn patches were combined into one boundary feature representing the fire as a whole. The result, WildfireBoundary (Figure 13), was visually confirmed to align with the historical location of the Devil's Head fire near Devils Head and Black Rock Mountain.

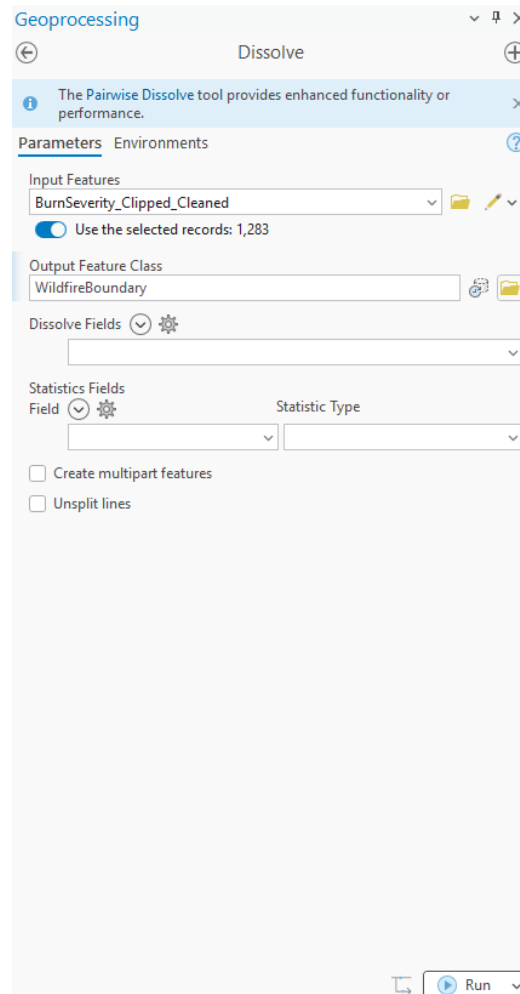


Figure 12. Dissolve tool parameters used to generate the WildfireBoundary feature class from the selected low/medium/high severity polygons (1,283 selected records).

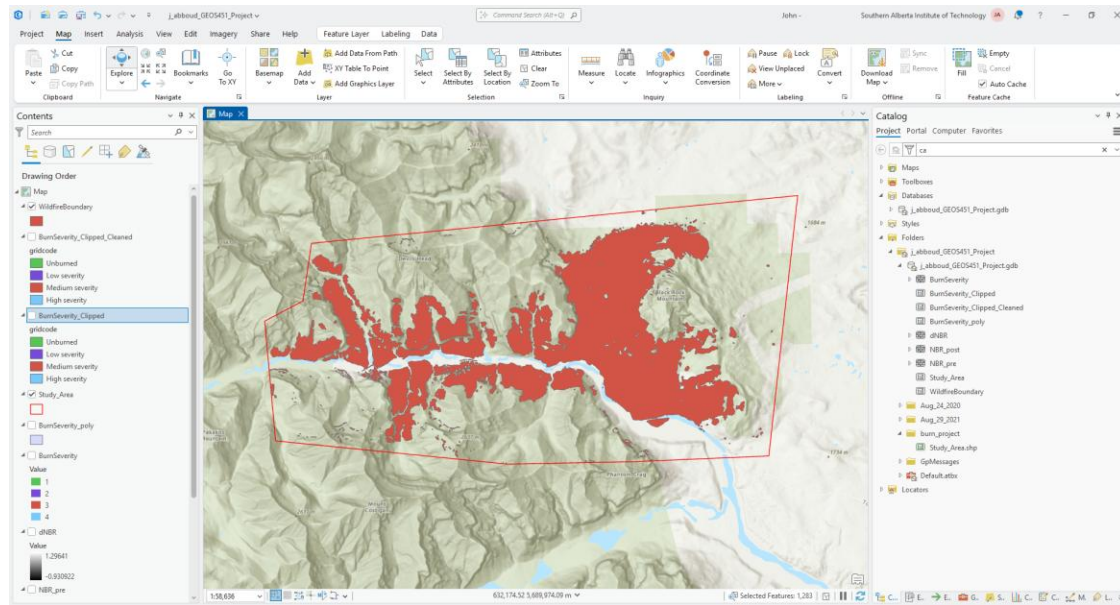


Figure 13. Final WildfireBoundary feature class, shown in relation to Devils Head and Black Rock Mountain.

3.7 Area Calculation

Prior to the final area calculations, the Study_Area boundary layer, originally supplied in NAD 1983 UTM Zone 11N, was reprojected to WGS 1984 UTM Zone 11N (EPSG:32611) using the Project tool, to match the coordinate system of the Sentinel-2 imagery and all derived analysis layers. BurnSeverity_poly was re-clipped against this reprojected boundary to produce the final BurnSeverity_Clippped_Cleaned and WildfireBoundary feature classes reported below. The reprojection changed the total burned area by approximately 0.06 hectares relative to the original NAD83-clipped result, a negligible difference consistent with the roughly 1 to 2 metre horizontal shift between the NAD83 and WGS84 datums in this area.

Area statistics were calculated using the Calculate Geometry tool (area in hectares) on both the WildfireBoundary and BurnSeverity_Clippped_Cleaned feature classes. For the severity feature class, the Summary Statistics tool was used to sum the area of all polygons within each severity class (grouped by the gridcode field), producing the table SeverityArea_Summary.

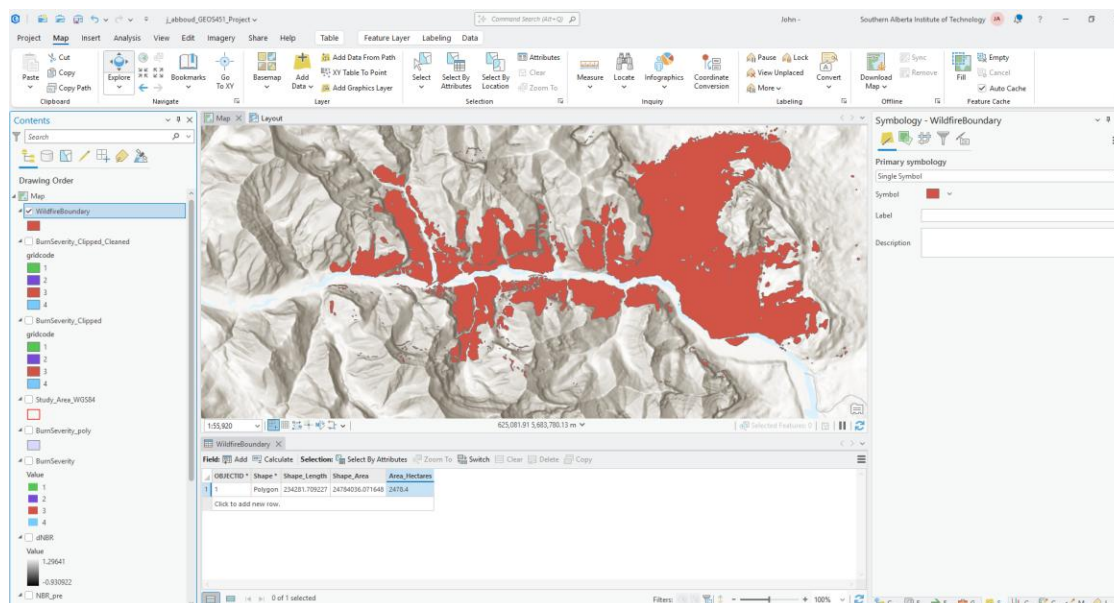


Figure 14. Calculate Geometry result for the WildfireBoundary feature class (post-reprojection to WGS 1984 UTM Zone 11N), giving a total burned area of 2,478.40 hectares.

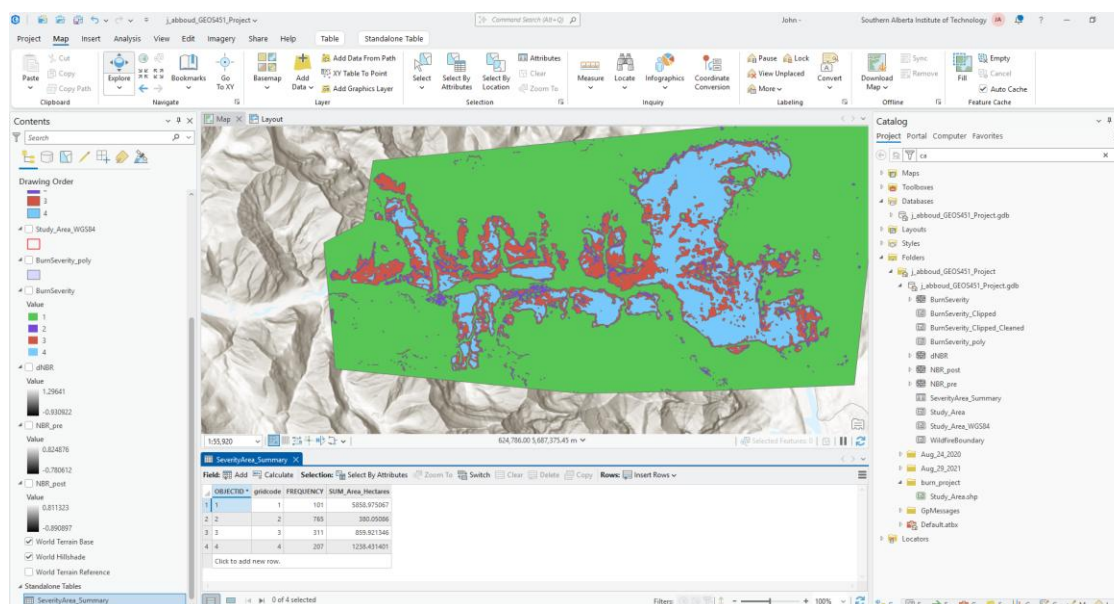


Figure 15. SeverityArea_Summary table (post-reprojection), giving total area (hectares) per severity class.

4. Results

The dNBR analysis delineated a total burned area of 2,478.40 hectares within the study area, closely matching the historically reported burned area of approximately 2,420 hectares for the Devil's Head wildfire (CBC News, 2021). The close agreement between the two figures, despite differences in mapping method, imagery resolution, and classification thresholds, supports the reliability of the dNBR approach used in this project.

Severity class	Area (hectares)	% of study area	% of burned area
Unburned	5,858.98	70.3%	—
Low severity	380.05	4.6%	15.3%
Medium severity	859.92	10.3%	34.7%
High severity	1,238.43	14.9%	50.0%
Total study area	8,337.38	100%	—
Total burned (low + medium + high)	2,478.40	29.7%	100%

High severity burn accounted for the largest share of the total burned area (50.0%), followed by medium severity (34.7%) and low severity (15.3%). This distribution, weighted toward high severity, is consistent with news coverage describing rapid, wind-driven fire growth (e.g., growth from 10 to 676 hectares over approximately three days at one point in the fire's progression), which typically produces a higher proportion of intense, high-severity burn compared to a slower-moving fire.

The total burned area calculated from the dissolved WildfireBoundary feature class (2,478.40 ha) matched exactly the sum of the low, medium, and high severity classes from the independently calculated SeverityArea_Summary table, confirming internal consistency between the two feature classes and indicating no loss or duplication of area between the boundary delineation and severity classification workflows.

5. Limitations and Discussion

- Temporal gap between pre- and post-fire imagery: because no cloud- and smoke-free Sentinel-2 scene was available in fall 2020, the post-fire image is approximately one year (rather than weeks or months) after the fire. This allowed roughly one growing season of vegetation regrowth within the burn scar, which likely causes a modest underestimation of severity, particularly at lower elevations and in areas with fast-recovering vegetation (e.g., grasses and shrubs), relative to a nearer-term post-fire image.
- Minimum mapping unit: the 400 m² noise-removal threshold, while necessary to remove classification speckle, may have merged a small number of genuinely small low-severity patches at the fire's margins into adjacent classes.
- Sensor resolution: Sentinel-2's 20 m native resolution in Band 12 (SWIR) limits the precision of the fire boundary and severity classification along narrow or highly fragmented burn edges relative to higher-resolution commercial imagery.

- Reclassification scheme: collapsing the standard six-class USGS/Key and Benson dNBR scheme into four classes (to match the project's low/medium/high requirement) necessarily simplifies the finer gradations of the original scheme; the specific class boundaries chosen (0.10, 0.27, 0.66) follow the original thresholds' moderate-low and moderate-high midpoints but represent a project-specific simplification rather than an independently validated threshold set.

Overall, despite these limitations, the close agreement between the calculated burned area (2,478.40 ha) and the independently reported historical figure (approximately 2,420 ha) suggests the method produced a reasonably accurate and defensible delineation of the Devil's Head wildfire's extent and severity.

References

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